

# SouthPol Exposure Time Calculator

*Work in progress*

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*Based on the spreadsheet from Antonio Mario Magalhaes (Copyright April 2026)*

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## 1 Introduction

This document presents an adaptation of the SouthPol exposure time calculator (ETC) developed by Prof. Antonio Mario Magalhaes (Copyright: April 2026) into a Python routine. The original ETC is a spreadsheet that calculates the compact photon-counting model for the SouthPol half-wave plate polarimeter. The  $f/5$  focal reducer has been assumed.

## 2 Initial parameters

The principal quantities considered are as follows:

- $Q$ : detector efficiency  $\times$  Atmosphere for the V band  $Q = 0.95 \cdot 0.85 = 0.81$  (FR not accounted; possibly  $\sim 0.96 = 0.995^8$  for a multi-layer, broadband coating on a 4-lens FR; TBD).
- $D$ : telescope diameter, set to 100 cm for SouthPol.
- $s$ : detector's scale, which is 0.28 arcsec/pix for SouthPol.
- $\Delta\lambda$ : filter bandwidth, set to 786 Å (for now).
- $\mu$ : sky brightness, set at 22 mag/arcsec<sup>2</sup>.
- $R_{\text{noise}}$ : detector reading noise, set to  $5e^-$  (for now).
- $N_{\text{plate}}$ : number of half-wave plate rotations, set to 8.
- $R_{\text{sat}}$ : saturation level of the detector, set to 60,000 (for now).
- Saturation limit: the default is 50,000  $e^-$ /pixel.

Input parameters:

- $m$ : stellar magnitude.
- $t$ : exposure time.
- FWHM: seeing, the default is 1.2 arcsec.
- $gain$ : detector's gain, the default is 0.5 arcsec.

| Typical Camera Performance With An STA 1600 CCD |                      |
|-------------------------------------------------|----------------------|
| Typical read noise 100kHz                       | 4.5e-                |
| Typical read noise 800kHz                       | 9.8e-                |
| Dark Current                                    | 0.001e-/pixel/sec    |
| Non-linearity                                   | <1%, 200e- to 100ke- |
| CCD size                                        | 95.04mm x 95.04mm    |
| CCD pixel size                                  | 9.0 $\mu$ m          |
| CCD pixel dimension                             | 10560 x 10560        |
| AR coatings available                           | Blue and Broadband   |

Figure 1: Detector parameters used in the calculation.

### 3 Derived parameters

- $N_{\text{pix}}$ : number of pixels covered by the stellar image.
- $dN_*/dt$ : stellar electron rate.
- $dN_{\text{sky}}/dt/\text{pix}$ : sky-background electron rate per pixel.
- $SNR$ : final signal-to-noise ratio.
- $\sigma_P$ : polarimetric precision in percent.

The formulas used in the original ETC are:

$$N_{\text{pix}} = \left( \frac{\text{FWHM}}{s} \right)^2, \quad (1)$$

$$\frac{dN_*}{dt} = Q \cdot 10^{-0.4(m+39.39)} \left( \frac{\pi}{4} \right) D^2 \Delta \lambda \frac{10^7}{3.6 \times 10^{-12}} \frac{1}{2}, \quad (2)$$

$$\frac{dN_{\text{sky}}}{dt \cdot \text{pix}} = Q \cdot 10^{-0.4(\mu+38.52)} \left( \frac{\pi}{4} \right) D^2 \Delta \lambda \frac{10^7}{3.6 \times 10^{-12}} s^2, \quad (3)$$

$$SNR = \frac{(dN_*/dt) \cdot t}{\sqrt{(dN_*/dt) \cdot t + 2N_{\text{pix}}[(dN_{\text{sky}}/dt/\text{pix}) \cdot t + R_{\text{noise}}^2]}}, \quad (4)$$

$$\sigma_P = \frac{\sqrt{N_{\text{plate}}}}{N_{\text{plate}}} \frac{100}{SNR} = \frac{100}{\sqrt{N_{\text{plate}}} SNR}. \quad (5)$$

### 4 Calculating the peak signal

Following a similar approach as in the Nordic Optical Telescope (NOT) ETC<sup>1</sup>, there are three important quantities useful for SouthPol:

- The total number of detected stellar electrons.
- The peak stellar electrons in one pixel for one exposure.
- The total peak counts (including the sky, converted to ADUs).

<sup>1</sup><https://www.not.iac.es/observing/forms/signal/index.php>

To calculate the above, we need to assume a two-dimensional Gaussian point-spread function (PSF). Additionally, we need to include in the SNR the object photon noise, the sky photon noise, and the read noise. In the case of multiple frames, average combining is assumed for the readout noise.

The detector reading properties assumed here for illustration purposes are based on the ALFOSC camera at NOT. The right parameters for SouthPol's new detector, such as the gain, the approximate full well in electrons/pixel, and the upper linearity limit in ADU/pixel, must be known and changed to obtain a more accurate result.

The dispersion of the Gaussian PSF is calculated as

$$\sigma_{\text{psf}} = \frac{FWHM}{2.355}. \quad (6)$$

The surface brightness at the center, or the peak stellar electrons, can be approximated by integrating over the area of the central pixel as

$$N_{\text{peak}}^* \approx N_{\text{total}}^* \frac{s^2}{2\pi\sigma_{\text{psf}}^2}, \quad (7)$$

where  $N_{\text{total}}^* = (dN_*/dt) \cdot t$  is the total stellar electrons in one exposure. For SouthPol, the peak should be evaluated per individual exposure, i.e., do not divide the peak by the number of sub-images. Finally, the total peak considering the sky contribution is

$$N_{\text{peak}}^{\text{total}} [e^-] = N_{\text{peak}}^* + N_{\text{sky}}, \quad (8)$$

or

$$N_{\text{peak}}^{\text{total}} [\text{ADU}] = \frac{N_{\text{peak}}^* + N_{\text{sky}}}{\text{gain}}, \quad (9)$$

with  $N_{\text{sky}} = (dN_{\text{sky}}/dt/\text{pix}) \cdot t$ .

Assuming a typical *gain* of 0.5 and a saturation limit of  $50000e^-$  per pixel, we can estimate if our calculation saturates and, if so, prompt a warning and calculate the minimum number of exposures needed to avoid saturation as

$$n_{\text{exposures}} = \frac{N_{\text{peak}}^{\text{total}}}{\text{saturation limit}}. \quad (10)$$

Figure 2 shows an example created for a SouthPol observation with the parameters explained above.

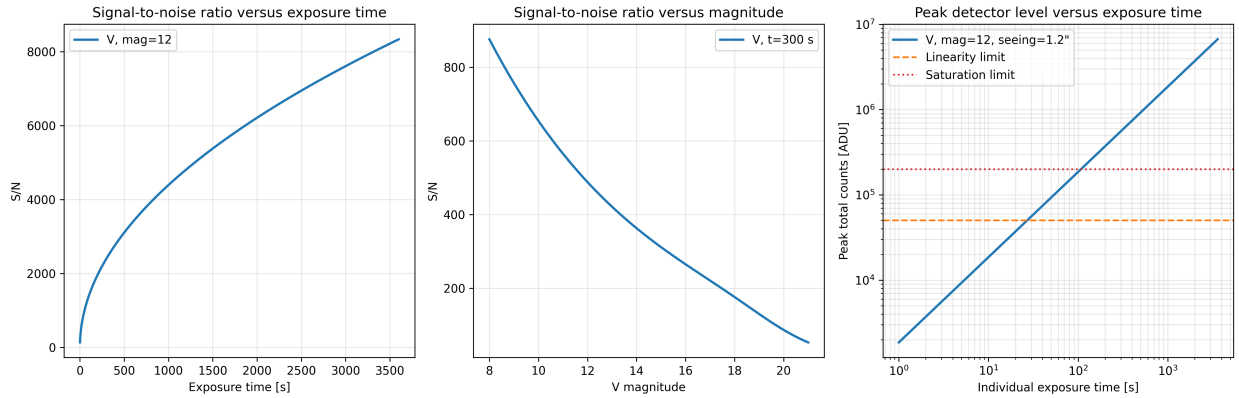


Figure 2: Example of an exposure time calculation for an observation down to V=12 mag, seeing of 1.2 and an exposure time of 300 s.